The Price of NOT on D4

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Revised from the 2026 preprint: the exact-cost appendix is simplified to a single interval-shell regime via the new unconditional lemma $W_f \subseteq B_f$ (Theorem 8), discovered during the machine-verified audit of the original proof. All headline results are machine-verified in Lean 4 (kernel axioms `propext`, `Quot.sound` at most; no `native_decide`, no `sorry`, no mathlib); see Section 9.

Abstract

This paper studies the Price of NOT on the resolved face of the four-state dual-rail carrier D4. Its sharp quantitative result is the exact resolved-face theorem $\nu(f) = \mathrm{dec}(f)$. The surrounding carrier material is used only as a compact ambient support package: the closed core $G = \mathrm{Clo}\{\wedge_k, \vee_k, P\}$, the confinement-breaking involution σ_s, the corrected P-equivariant-collapse theorem, and the absorbed-face transfer needed to place the exact theorem inside one compatible carrier presentation. Appendix D contains the proof architecture for the exact-cost theorem — in this edition a single interval-shell regime, made unconditional by the new lemma $W_f \subseteq B_f$.

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1 Introduction

This paper starts from the earlier resolved-block theorem of Paper I [8] and asks whether its exact price of negation persists when viewed through a compatible four-state dual-rail carrier frame D4. It does: on the resolved block $R = \{\mathsf{FAL}, \mathsf{TRU}\}$, the minimal syntactic negation count $\nu(f)$ is exactly the chain decrease number $\mathrm{dec}(f)$ for every nonconstant $f : R^n \to R$. The larger carrier frame is admitted only insofar as it sharpens the boundary around this exact law.

For Boolean functions, the price of negation is classical: Markov [2] determined the circuit inversion complexity as $\lceil \log_2(d(f)+1) \rceil$, and Morizumi [3] proved the formula-level law is exact, $I_{\mathrm{for}}(f) = d(f)$. The present paper proves the formula-exact law in the four-valued dual-rail setting, with a proof intrinsic to the recovered resolved-face language; the relation to the Boolean transport route is discussed at Theorem 4. For the surrounding negation-limited literature — the Markov-style upper-bound argument [9], constant-depth circuits [10], bounded-depth circuits [11], lower bounds against few negations [12], negation-limited formula structure [13], and multi-valued circuit-model laws [14] — the formula-exact multi-valued law proved here appears to be new.

The supporting package has six layers: (1) the $\mathsf{D4}$ carrier and its two-coordinate presentation; (2) the qualitative closed core $G = \mathrm{Clo}\{\wedge_k, \vee_k, P\}$; (3) the confinement-breaking operation σ_s; (4) the corrected expressive-collapse theorem; (5) the gate-absorbed resolved-face transfer into Theorem 4; (6) the unary geometric and narrow execution specializations.

The paper does not claim that adjoining σ_s yields all functions on $\mathsf{D4}$; the correct target class is the full P-equivariant clone. It also does not promote NCP, the Arrow of Computation, variable-cost RG, or empirical bridge results to headline theorems. The outer crown/constant-barrier picture belongs more naturally to a broader carrier companion line; here it remains background, while $\nu(f) = \mathrm{dec}(f)$ remains the sharp quantitative face of the paper.

The closed core preserves the internal symmetry and endpoint discipline of the carrier, while σ_s breaks that discipline without crossing the rail-swap fence. On the resolved face, the earlier theorem surface is recovered not as a plain restriction $G|_R$, but through a fixed absorbed-face presentation compatible with the decompressed carrier theory. This supplies an ambient gateway into the exact theorem rather than a replacement theorem spine. The appendix structure mirrors that proof architecture: Appendix A records the closed-core background, Appendix B proves the corrected collapse, Appendix C fixes the absorbed-face transfer, Appendix D contains the exact-cost proof, and Appendix E isolates the chain bookkeeping used there.

What changed in this edition. The audit of the original exact-cost appendix (a full machine replay of every layer) revealed that its residual regime — the local witness reduction, star-local fiber coherence, and five-case rerouting layers — is vacuous: the containment $W_f \subseteq B_f$ holds *unconditionally* (Theorem 8), so the interval-shell regime always applies and the residual layers can never be reached. This edition therefore proves the upper bound through the single interval-shell regime. The original compressed classification arguments are superseded, not corrected: no error was found in any reachable layer.

2 The $\mathsf{D4}$ Carrier and the Closed Monotone Core

2.1 Carrier

Let $\mathsf{D4} = \{\mathsf{UNK}, \mathsf{FAL}, \mathsf{TRU}, \mathsf{CON}\}$, identified with the dual-rail carrier $\{0,1\}^2$ by

$$\mathsf{UNK} = (0,0), \quad \mathsf{FAL} = (0,1), \quad \mathsf{TRU} = (1,0), \quad \mathsf{CON} = (1,1).$$

Use the coordinate functions $s = (e^+ + e^-)/2$ and $d = (e^+ - e^-)/2$, so that $s(\mathsf{UNK}) = 0$, $s(\mathsf{FAL}) = s(\mathsf{TRU}) = 1/2$, $s(\mathsf{CON}) = 1$, and $d(\mathsf{UNK}) = 0$, $d(\mathsf{FAL}) = -1/2$, $d(\mathsf{TRU}) = 1/2$, $d(\mathsf{CON}) = 0$.

Let P denote rail-swap: $P(\mathsf{UNK}) = \mathsf{UNK}$, $P(\mathsf{FAL}) = \mathsf{TRU}$, $P(\mathsf{TRU}) = \mathsf{FAL}$, $P(\mathsf{CON}) = \mathsf{CON}$. The internal lattice operations are coordinatewise meet and join in the knowledge order:

$$(a,b) \wedge_k (c,d) = (a \wedge c,\, b \wedge d), \qquad (a,b) \vee_k (c,d) = (a \vee c,\, b \vee d).$$

2.2 Closed Core

Define the closed carrier core $G = \mathrm{Clo}\{\wedge_k, \vee_k, P\}$. In the main text we use only the following qualitative theorem.

Theorem 1 (Qualitative closed-core soundness)**.** *Every term operation in G is monotone for the knowledge order, preserves both endpoints* UNK *and* CON*, and is P-equivariant.*

This is the forward-confinement background for the rest of the paper. Stronger converse characterizations are left to Appendices A and B.

3 The Confinement-Breaking Operation

Define the simultaneous rail-complement involution $\sigma_s(a, b) = (1 - a, 1 - b)$; equivalently $\sigma_s(\mathsf{UNK}) = \mathsf{CON}$, $\sigma_s(\mathsf{FAL}) = \mathsf{TRU}$, $\sigma_s(\mathsf{TRU}) = \mathsf{FAL}$, $\sigma_s(\mathsf{CON}) = \mathsf{UNK}$.

Theorem 2 (Confinement-breaking algebraic move)**.** *The operation σ_s is an involution, commutes with P, fails to preserve both endpoints, and is non-monotone for the knowledge order.*

These are the only carrier-level facts needed later. No uniqueness claim is used in the main theorem surface; when a stronger phrase is convenient, the safe one is that σ_s is a non-monotone completing primitive.

4 Carrier Collapse with the Correct Symmetry Fence

For each arity $n \geq 1$, let C_n be the n-ary term clone generated by projections together with $\wedge_k, \vee_k, P, \sigma_s$. Let P act coordinatewise on $\mathsf{D4}^n$, and define

$$\mathrm{Eqv}_P(n) = \{f : \mathsf{D4}^n \to \mathsf{D4} \mid f(Px) = P(f(x)) \text{ for all } x\}.$$

Theorem 3 (Corrected expressive-collapse theorem)**.** *For every arity $n \geq 1$, $C_n = \mathrm{Eqv}_P(n)$.*

Write $x_i = (p_i, q_i)$ and $bits(x) = (p_1, q_1, \ldots, p_n, q_n)$, and let S denote rail-swap on bit tuples.

Lemma 1 (Rail-swap soundness)**.** *For every n-ary term t built from projections, $\wedge_k$, $\vee_k$, P, and σ_s, there exists a Boolean function $\phi_t : \{0, 1\}^{2n} \to \{0, 1\}$ such that $t(x) = (\phi_t(bits(x)), \phi_t(S(bits(x))))$ for all $x \in \mathsf{D4}^n$. In particular, every term operation in C_n is P-equivariant.*

Lemma 2 (Formula lifting)**.** *Let ϕ be a Boolean formula in the variables $p_1, q_1, \ldots, p_n, q_n$. Then there exists an n-ary carrier term T_ϕ over $\wedge_k, \vee_k, P, \sigma_s$ such that $T_\phi(x) = (\phi(bits(x)), \phi(S(bits(x))))$ for all $x \in \mathsf{D4}^n$.*

Theorem 3 follows by combining Lemma 1 with Lemma 2: the generators enforce exactly P-equivariance, and every P-equivariant carrier map is recovered from a single Boolean coordinate formula; the full proof is given in Appendix B.

Two boundary remarks will be used later. Adjoining σ_s yields the maximal class compatible with rail-swap symmetry, namely the full P-equivariant clone, not all carrier maps; compare the finite-set clone viewpoint in [6, 7]. On the resolved face $R = \{\mathsf{FAL}, \mathsf{TRU}\}$, the zero-constant outer fragment is the P-equivariant crown fragment, and adjoining one resolved constant completes it. This gives a coarse outer invariant ν_{carrier}, but not the paper's main quantitative theorem.

5 Resolved-Face Transfer and Exact Cost

Let $R = \{\mathsf{FAL}, \mathsf{TRU}\}$. The resolved-face theory is not obtained by plain restriction of the internal carrier generators.

Proposition 1 (Plain restriction is insufficient)**.** *On mixed resolved inputs, the internal carrier operations leave the resolved face:*

$$\wedge_k(\mathsf{FAL},\mathsf{TRU})=\mathsf{UNK},\quad \wedge_k(\mathsf{TRU},\mathsf{FAL})=\mathsf{UNK},\quad \vee_k(\mathsf{FAL},\mathsf{TRU})=\mathsf{CON},\quad \vee_k(\mathsf{TRU},\mathsf{FAL})=\mathsf{CON}.$$

Hence the resolved-block operations of the narrow manuscript cannot be obtained by plain restriction of the internal carrier generators.

Fix designated re-entry data $\rho_{\text{meet}}(\mathsf{UNK})=\mathsf{FAL}$, $\rho_{\text{join}}(\mathsf{CON})=\mathsf{TRU}$, with $\rho_{\text{meet}}|_R=\rho_{\text{join}}|_R=\mathrm{id}_R$, and define

$$\text{meet}_R(a,b)=\rho_{\text{meet}}(a\wedge_k b),\qquad \text{join}_R(a,b)=\rho_{\text{join}}(a\vee_k b).$$

Proposition 2 (Gate-absorbed recovery of the resolved face)**.** *After fixing the designated re-entry data, the resolved-face operations are recovered from carrier evaluation by absorbed re-entry on the only escaping mixed cases.*

Transfer principle. The earlier resolved-block language is not obtained by plain restriction of $\wedge_k,\vee_k,P$ to R; rather, after fixing the designated re-entry data for the two escaping mixed cases, it is recovered through a specific absorbed-face presentation compatible with the decompressed carrier theory. We use this presentation only as the gateway from carrier background to the exact resolved-face theorem. The underlying table lemmas are recorded in Appendix C.

The exact-cost proof is phrased in order-theoretic terms: $\mathrm{mcr}(f)$ is the maximal chain-reversal count and $\nu(f)$ is the resolved-face negation complexity in the recovered term language. An alternative Boolean transport route may be compared with Morizumi's formula-level inversion result [3], but the proof used here is intrinsic to the recovered resolved-face language.

Theorem 4 (Exact negation complexity on the resolved face)**.** *For every nonconstant function $f:R^n\to R$, the minimal number $\nu(f)$ of syntactic negation occurrences is exactly the chain decrease number* $\mathrm{dec}(f)$*.*

Proof. The lower bound $\mathrm{mcr}(f)\le\nu(f)$ is Proposition 10: projections contribute no chain reversal (Lemma 3), meet_R and join_R are reversal-subadditive (Lemma 4), and one occurrence of σ contributes at most one new reversal along any chain (Lemma 5). For the upper bound, Theorem 8 gives the containment $W_f\subseteq B_f$ unconditionally, so the canonical shell $S_f=A_f\cap{\downarrow}W_f$ is an A_f-ideal with $W_f\subseteq S_f\subseteq B_f$ (Corollary 5); Proposition 11 realizes χ_{S_f} with one occurrence of σ, Theorem 7 gives $\mathrm{mcr}(f_{S_f})=\mathrm{mcr}(f)-1$, and the induction of Theorem 9 yields $\nu(f)\le\mathrm{mcr}(f)$. The exact-equality corollary of Appendix D assembles $\nu(f)=\mathrm{mcr}(f)=\mathrm{dec}(f)$. Every cited result is proved in full in Appendix D. □

This remains the sharp observable of the paper; the larger carrier story is used only as compact ambient support for this theorem and should not be read as displacing it. Appendices B and C prepare the recovered resolved face, Appendix D proves the exact theorem, and Appendix E isolates the auxiliary chain language used in that proof.

6 Unary Geometry of Negation

On the resolved face $R=\{\mathsf{FAL},\mathsf{TRU}\}$, write $\sigma=\sigma_s|_R$. Since both P and σ_s exchange FAL and TRU, their restrictions to R agree.

Proposition 3 (Unary resolved-face term classification)**.** *The unary resolved-face term functions are exactly*

$$\mathrm{id}_R, \qquad \sigma, \qquad c_{\mathsf{FAL}}(x) = \mathrm{meet}_R(x, \sigma(x)), \qquad c_{\mathsf{TRU}}(x) = \mathrm{join}_R(x, \sigma(x)).$$

Theorem 5 (Unary flip-localization)**.** *In the unary dual-rail resolved slice, the restricted negation $\sigma = \sigma_s|_R = P|_R$ is the unique flip-sign reverser.*

This is the cleanest geometric specialization of the exact-cost theorem: in the unary resolved slice, σ is the only genuine direction-reversing map.

7 Executed Reversals Under Finite Episode Energy

The execution layer remains narrow. It specializes the finite-event budget argument of Paper I [8] and does not claim a thermodynamic theorem in the sense of the classical Landauer–Bennett discussion [5, 4]. We count only executed unary resolved reversals, namely events moving between FAL and TRU on the resolved slice.

Theorem 6 (Finite executed reversal bound)**.** *Under finite episode energy $E_\tau < \infty$ and a fixed positive cost $\epsilon_{\mathrm{rev}} > 0$ per executed unary flip-sign reversal, only finitely many such executed reversals can occur, bounded by $\lfloor E_\tau / \epsilon_{\mathrm{rev}} \rfloor$.*

Corollary 1 (Bounded reversal reachability)**.** *A finite episode may fail to realize some unary resolved reversals that become realizable only in a larger episode with a larger available budget.*

This section does not claim a variable-cost RG theorem, a three-channel energy theorem, a general syntax-versus-execution lower bound, or a higher-arity execution classification.

8 Scope and Outlook

The paper proves a broader theorem package than the earlier resolved-block manuscript, but the exact observable remains $\nu(f) = \mathrm{dec}(f)$ on the resolved face. Carrier-level additions are admitted only if they sharpen that observable. For the same reason, the broader carrier line is companion background rather than the main ownership claim of the present paper. The following remain outside the main theorem surface: NCP as a headline theorem; the Arrow of Computation as a headline theorem; variable-cost RG or three-channel energy theoremization; empirical bridge material.

8.1 Appendix Order

The appendix order is: Appendix A, qualitative and enumerative proof material for G; Appendix B, the full proof of corrected Theorem 3; Appendix C, the decompression and transfer lemmas placing the resolved face correctly; Appendix D, the exact-complexity proof package for Theorem 4 (single interval-shell regime in this edition); Appendix E, chain-reversal, mcr, and Λ^{up} tools where they clarify the exact theorem.

9 Mechanization

Every headline statement of this paper is machine-verified in dependency-minimal Lean 4 (core only; no mathlib; no `native_decide`; no `sorry`; kernel axiom profile at most `[propext, Quot.sound]`, with the finite table facts axiom-free). The artifact accompanies this paper and comprises:

- **Theorem 1** — `cterm_monotone`, `cterm_endpoints`, `cterm_equivariant` (structural induction over a closed-core term syntax);
- **Theorem 2** — `sigd_invol`, `sigd_Pd`, `sigd_breaks_endpoints`, `sigma_not_monotone` (axiom-free);
- **Propositions C.1/C.2/C.4** — `escape`, `recovery`, `reentry_independent` (axiom-free);
- **Appendix D lower bound** — the per-chain subadditivity lemmas `chainDrops_and`, `chainDrops_or`, and the end-value-refined alternation law `chainDrops_not`, assembled into `dec_le_sigCount`;
- **Appendix D upper bound** — the nested one-σ peel is built on a mechanized tower library: the interval-shell decrement, the explicit uniform witness, and the constructive monotone extension (`tower_of_decPts`), together with the monotone-DNF realization (`mono_realize`) and the tower-to-term assembly (`tower_realize`);
- **Theorem 4** — `nu_eq_dec_D4`: the exact equality, stated with terms evaluated on the resolved face by the absorbed operations meet_R, join_R and σ, for nonconstant f on the n-cube.

In addition, the original appendix was audited by exhaustive machine replay: the closed-core and collapse layers, the transfer tables, the interval-shell theorem (all 464 valid shells at $n \leq 3$; the canonical shell at $n = 4$), and the inductive completion (65,368/65,368 nonconstant cases at $n \leq 4$). The audit also verified computationally that the residual of the interval regime is empty for all $n \leq 4$ — the observation that led to Theorem 8. The replay scripts and logs are included in the artifact repository.

A Qualitative and Enumerative Core Material

This appendix records the forward qualitative facts about the closed core $G = \text{Clo}\{\wedge_k, \vee_k, P\}$ used in the main text. Their broader natural home is the companion carrier line; here they are retained only to keep the transfer-to-exact package self-contained.

A.1 Proof of the qualitative soundness theorem

Proof of Theorem 1. Each generator is monotone for the knowledge order. The operations $\wedge_k$ and $\vee_k$ are coordinatewise meet and join on $\{0,1\}^2$, hence are monotone. The involution P is order-preserving because it only swaps the two coordinates.

Each generator also preserves the endpoints $\mathsf{UNK} = (0,0)$ and $\mathsf{CON} = (1,1)$: $\wedge_k(\mathsf{UNK},\mathsf{UNK}) = \mathsf{UNK}$, $\vee_k(\mathsf{UNK},\mathsf{UNK}) = \mathsf{UNK}$, $\wedge_k(\mathsf{CON},\mathsf{CON}) = \mathsf{CON}$, $\vee_k(\mathsf{CON},\mathsf{CON}) = \mathsf{CON}$, and $P(\mathsf{UNK}) = \mathsf{UNK}$, $P(\mathsf{CON}) = \mathsf{CON}$.

Finally, every generator is P-equivariant. For the involution itself this is immediate. For $\wedge_k$ and $\vee_k$, coordinatewise meet and join commute with swapping the two coordinates. Since monotonicity, endpoint preservation, and P-equivariance are all stable under composition, every term operation in G has the same three properties. $\square$

A.2 Small carrier checks

The proof above uses only the generator-level identities. For convenience, we record the two endpoint rows explicitly:

$$\mathsf{UNK} \wedge_k x = \mathsf{UNK}, \quad \mathsf{UNK} \vee_k x = x, \qquad \mathsf{CON} \wedge_k x = x, \quad \mathsf{CON} \vee_k x = \mathsf{CON}.$$

Thus UNK and CON behave as the lower and upper endpoints of the internal knowledge order, while P fixes those endpoints and exchanges the resolved states FAL and TRU.

Remark. The appendix does not attempt a converse description of the full core G. The main text uses only the forward confinement theorem, and the stronger carrier-collapse statement appears only after adjoining σ_s in Appendix B.

A.3 Orbit detectors on the resolved face

Work now on the resolved face $R = \{\mathsf{FAL}, \mathsf{TRU}\}$, $E = \{\mathsf{UNK}, \mathsf{CON}\}$. Every P-orbit in R^n has size 2, and each orbit has a unique representative whose first coordinate is FAL.

Proposition 4. *For $i \neq j$, define*

$$\begin{aligned} Eq_{ij}(x) &= \vee_k\big(\wedge_k(x_i, x_j),\ \wedge_k(P(x_i), P(x_j))\big), \\ Neq_{ij}(x) &= \vee_k\big(\wedge_k(x_i, P(x_j)),\ \wedge_k(P(x_i), x_j)\big). \end{aligned}$$

On R^n, both maps are E-valued. Moreover, $Eq_{ij}(x) = \mathsf{CON} \iff x_i = x_j$ and $Neq_{ij}(x) = \mathsf{CON} \iff x_i \neq x_j$.

Proof. Check the four resolved input pairs. If $x_i = x_j$, then exactly one of the two inner meets in the definition of Eq_{ij} contributes a resolved value and the other contributes its P-partner, so the outer join is CON; otherwise both inner meets are UNK. The same calculation with one rail swapped gives the statement for Neq_{ij}. $\square$

For $\alpha \in \{0,1\}^{n-1}$, define $D_\alpha(x) = \bigwedge_{j=2}^{n} B_j(x)$, where $B_j = Eq_{1j}$ if $\alpha_{j-1} = 0$ and $B_j = Neq_{1j}$ if $\alpha_{j-1} = 1$. For $A \subseteq \{0,1\}^{n-1}$, set $D_A = \bigvee_{\alpha \in A} D_\alpha$.

Proposition 5. *Each D_α is E-valued and satisfies $D_\alpha(x) = \mathsf{CON}$ exactly on the P-orbit indexed by α, while $D_\alpha(x) = \mathsf{UNK}$ on all other orbits.*

Proof. By Proposition 4, each factor B_j detects whether the j-th coordinate agrees or disagrees with the first coordinate exactly as prescribed by α. Since the factors are E-valued, their knowledge-meet acts as conjunction on these orbit conditions. Hence $D_\alpha = \mathsf{CON}$ exactly on the orbit with pattern α. $\square$

A.4 The outer resolved-face crown

Remark. The broader carrier companion line proves the following resolved-face crown fact: before adjoining any resolved constant, the R-valued restrictions of G on R^n are exactly the P-equivariant maps $h : R^n \to R$, and their count is $2^{2^{n-1}}$. In the present paper we use only the qualitative boundary consequence that the zero-constant outer fragment remains inside the P-equivariant crown.

A.5 One resolved constant completes the outer fragment

Remark. The same companion line also proves that adjoining one resolved constant completes the outer resolved-face fragment to all maps $R^n \to R$. The present paper uses only the resulting outer zero/one accessibility contrast, not the full theorem-proof package itself. This binary outer boundary is kept only as context around the finer exact theorem $\nu(f) = \mathrm{dec}(f)$.

B Proof of the Corrected Expressive-Collapse Theorem

This appendix expands the proof of Theorem 3 only to the extent needed for the present paper's self-contained carrier-to-resolved-face gateway.

B.1 Rail-swap normal form

Proof of Lemma 1. Proceed by induction on the term t. For a projection $t(x) = x_i = (p_i, q_i)$, take $\phi_t(\mathit{bits}(x)) = p_i$; then $t(x) = (\phi_t(\mathit{bits}(x)), \phi_t(S(\mathit{bits}(x))))$.

If the claim holds for u and v, it is preserved by $\wedge_k$ and $\vee_k$ because these are coordinatewise Boolean conjunction and disjunction on $\{0,1\}^2$. Hence the first coordinate of $u \wedge_k v$ is $\phi_u \wedge \phi_v$, and the first coordinate of $u \vee_k v$ is $\phi_u \vee \phi_v$.

If the claim holds for u, then it also holds for $P(u)$ because applying P interchanges the two coordinates: $P(u)(x) = (\phi_u(S(\mathit{bits}(x))), \phi_u(\mathit{bits}(x)))$. Thus one may take $\phi_{P(u)} = \phi_u \circ S$.

If the claim holds for u, then it also holds for $\sigma_s(u)$ because $\sigma_s(a, b) = (1 - a, 1 - b)$: $\sigma_s(u)(x) = (1 - \phi_u(\mathit{bits}(x)),\ 1 - \phi_u(S(\mathit{bits}(x))))$. Hence one may take $\phi_{\sigma_s(u)} = \neg\phi_u$.

This completes the induction. In particular, every term operation obtained from the generators is P-equivariant. □

B.2 Formula lifting

Proof of Lemma 2. Translate Boolean formulas recursively into carrier terms. For a positive rail variable p_i, use the carrier variable x_i itself. For a negative rail variable q_i, use $P(x_i)$, whose first coordinate is q_i. For conjunction and disjunction, use $\wedge_k$ and $\vee_k$. For negation, use σ_s. More precisely, define a term T_ϕ by recursion on ϕ:

$$T_{p_i} = x_i, \quad T_{q_i} = P(x_i), \quad T_{\phi\wedge\psi} = T_\phi \wedge_k T_\psi, \quad T_{\phi\vee\psi} = T_\phi \vee_k T_\psi, \quad T_{\neg\phi} = \sigma_s(T_\phi).$$

An induction on the Boolean formula now shows that the first coordinate of $T_\phi(x)$ is exactly $\phi(\mathit{bits}(x))$, while the second coordinate is the same formula evaluated on the swapped bit tuple $S(\mathit{bits}(x))$. □

B.3 Collapse to the full P-equivariant clone

Proof of Theorem 3. The inclusion $C_n \subseteq \mathrm{Eqv}_P(n)$ is exactly the final sentence of Lemma 1. For the converse, let $f \in \mathrm{Eqv}_P(n)$. Define the Boolean coordinate function $\phi_f(\mathit{bits}(x)) = \pi_1(f(x))$, where π_1 denotes the first rail coordinate. Since f is P-equivariant, the second rail coordinate is determined by the same function on the swapped tuple: $f(x) = (\phi_f(\mathit{bits}(x)),\ \phi_f(S(\mathit{bits}(x))))$. Lemma 2 therefore produces a carrier term T_{ϕ_f} with exactly this value, so $f \in C_n$. □

C Decompression and Transfer Lemmas

This appendix records the table-level lemmas behind the absorbed recovered resolved-face operations.

C.1 Escaping mixed cases

Proposition 6. *For $a, b \in R = \{\mathsf{FAL}, \mathsf{TRU}\}$, the internal carrier operations stay in R except on the mixed inputs, where they escape in exactly one way: $\wedge_k(\mathsf{FAL}, \mathsf{TRU}) = \wedge_k(\mathsf{TRU}, \mathsf{FAL}) = \mathsf{UNK}$ and $\vee_k(\mathsf{FAL}, \mathsf{TRU}) = \vee_k(\mathsf{TRU}, \mathsf{FAL}) = \mathsf{CON}$.*

Proof. This is an immediate coordinatewise calculation in the dual-rail model $\mathsf{FAL} = (0, 1)$, $\mathsf{TRU} = (1, 0)$. Taking coordinatewise meet produces $(0, 0) = \mathsf{UNK}$, while taking coordinatewise join produces $(1, 1) = \mathsf{CON}$. On the unmixed inputs, one obtains the original resolved state again. □

C.2 Recovered resolved-face operations

Proposition 7. *With the designated re-entry data $\rho_{\text{meet}}(\mathsf{UNK}) = \mathsf{FAL}$, $\rho_{\text{join}}(\mathsf{CON}) = \mathsf{TRU}$, and $\rho_{\text{meet}}|_R = \rho_{\text{join}}|_R = \mathrm{id}_R$, the absorbed recovered operations satisfy $\mathrm{meet}_R(a, b) = \min(a, b)$ and $\mathrm{join}_R(a, b) = \max(a, b)$ for all $a, b \in R$.*

Proof. On unmixed pairs, Proposition 6 shows that the internal operation already lands in R, so the re-entry maps act trivially. On the two mixed pairs, the meet case lands in UNK and is sent back to FAL, while the join case lands in CON and is sent back to TRU. This is exactly the two-point chain minimum and maximum on $R = \{\mathsf{FAL}, \mathsf{TRU}\}$. □

Corollary 2. *The resolved-face term language used in Section 5 is not the plain restriction of the internal carrier generators. It is the absorbed R-face obtained by evaluating $\wedge_k$ and $\vee_k$ in the carrier and then applying the designated re-entry data on the two escaping mixed cases.*

Proof. The failure of plain restriction is Proposition 1. The recovered operations are Proposition 7. □

C.3 Witness-independence at the resolved endpoints

Proposition 8. *Suppose ρ'_{meet} and ρ'_{join} are alternative designated re-entry maps with the same endpoint values: $\rho'_{\text{meet}}|_R = \mathrm{id}_R$, $\rho'_{\text{meet}}(\mathsf{UNK}) = \mathsf{FAL}$, $\rho'_{\text{join}}|_R = \mathrm{id}_R$, $\rho'_{\text{join}}(\mathsf{CON}) = \mathsf{TRU}$. Then the induced binary operations on R agree with meet_R and join_R on all inputs from R^2.*

Proof. Proposition 6 shows that the only possible outputs of $\wedge_k$ on R^2 are $\mathsf{FAL}, \mathsf{TRU}, \mathsf{UNK}$, and the only possible outputs of $\vee_k$ on R^2 are $\mathsf{FAL}, \mathsf{TRU}, \mathsf{CON}$. The two re-entry systems agree on exactly these values. Hence they induce identical binary operations on the resolved face. □

Remark. This extensional agreement does not imply uniqueness of the surrounding carrier realization. Different decompressed witness systems may still behave differently away from the endpoint values used here. The paper does not count or classify that ambient witness dependence.

C.4 Transport to the recovered resolved-face language

Proposition 9. *Let $\sigma = P|_R$. Then the term language generated on R by* meet_R, join_R, σ *is exactly the recovered resolved-face language used in Section 5.*

Proof. Proposition 7 identifies meet_R and join_R with the two-point minimum and maximum on $R = \{\mathsf{FAL}, \mathsf{TRU}\}$. The unary map $\sigma = P|_R$ exchanges the two resolved endpoints. These are precisely the operations used in the recovered resolved-face term algebra. □

Remark (Safe transfer principle). The appendix proves only the narrow statement needed in the main text: the resolved-face language is not obtained by plain restriction, but by absorbed re-entry on the two escaping mixed cases. It does not prove a quotient theorem, a congruence-image theorem, or any theorem counting ambient witness use.

D Exact Negation Complexity on the Resolved Face

This appendix gives the proof of Theorem 4. In this edition the exact-cost argument has three layers: lower-bound charging, the interval-shell decrement made unconditional by the new lemma $W_f \subseteq B_f$, and inductive completion.

Comparison with the original preprint. The original appendix carried three further layers — local witness reduction, star-local fiber coherence, and a five-case rerouting package — to handle a residual family of shells assumed to fall outside the interval regime. Theorem 8 below shows that this residual family is empty: the containment $W_f \subseteq B_f$ holds for every nonconstant f, so the interval-shell regime already covers every inductive step. The removed layers were never reachable; no error in them was implicated. This simplification was found during the machine-verified audit of the appendix.

D.1 Lower-Bound Charging

Let $R = \{\mathsf{FAL}, \mathsf{TRU}\}$ with the chain order $\mathsf{FAL} < \mathsf{TRU}$, and let R^n carry the componentwise order. For a maximal chain $C = (c_0 < \cdots < c_n)$ define $\mathrm{desc}(f, C)$ to be the number of indices i such that $f(c_i) = \mathsf{TRU}$, $f(c_{i+1}) = \mathsf{FAL}$. Set $\mathrm{mcr}(f) = \max_C \mathrm{desc}(f, C)$. On the two-point chain one has $\mathrm{mcr}(f) = \mathrm{dec}(f)$.

Lemma 3. *If t is a projection, then* $\mathrm{desc}(t, C) = 0$ *for every maximal chain C.*

Proof. A projection is monotone on R^n, so along any increasing chain it can change only from FAL to TRU, never from TRU to FAL. □

Lemma 4. *For resolved-face terms u and v, and every maximal chain C,*

$$\mathrm{desc}(\mathrm{meet}_R(u, v), C) \le \mathrm{desc}(u, C) + \mathrm{desc}(v, C), \qquad \mathrm{desc}(\mathrm{join}_R(u, v), C) \le \mathrm{desc}(u, C) + \mathrm{desc}(v, C).$$

Proof. If $\mathrm{meet}_R(u, v)$ descends at a step of C, then both inputs are TRU below and at least one is FAL above, so that descent is charged to an input descent. If $\mathrm{join}_R(u, v)$ descends, then both inputs are FAL above and at least one is TRU below, so again the descent is charged to an input descent. Summing over all steps gives the inequalities. □

Lemma 5. *For every resolved-face term u and maximal chain C,* $\mathrm{desc}(\sigma(u), C) \le \mathrm{desc}(u, C) + 1$*.*

Proof. Along a fixed chain, applying σ exchanges FAL and TRU, so descents of $\sigma(u)$ are ascents of u. In any binary word, the number of ascents exceeds the number of descents by at most one. $\square$

Proposition 10 (Lower-bound charging). *If a resolved-face term t computes $f : R^n \to R$, then* $\mathrm{mcr}(f) \le \nu(t)$. *In particular,* $\mathrm{dec}(f) = \mathrm{mcr}(f) \le \nu(f)$.

Proof. Proceed by structural induction on t. Lemma 3 handles projections. Lemma 4 handles meet_R and join_R. Lemma 5 handles one occurrence of σ, which can add at most one new descent along any chain. Therefore any term computing f has at least $\mathrm{mcr}(f)$ occurrences of σ. Taking the minimum over all such terms gives $\mathrm{mcr}(f) \le \nu(f)$, and $\mathrm{mcr}(f) = \mathrm{dec}(f)$ on the two-point chain. $\square$

D.2 Interval-Shell Decrement

Assume now that $f : R^n \to R$ is nonconstant and non-monotone. Define

$$A_f = \{x : f(x) = \mathsf{TRU}\}, \qquad G_f = {\uparrow}A_f \setminus A_f, \qquad B_f = A_f \setminus {\uparrow}G_f.$$

Let W_f be the union of the first TRU-blocks over all worst chains of f. For $S \subseteq R^n$, define the lowered residual f_S by $f_S(x) = \mathsf{FAL}$ for $x \in S$, $f_S(x) = f(x)$ otherwise.

Proposition 11 (One-σ realization for interval shells). *Let S be an A_f-ideal with $W_f \subseteq S \subseteq B_f$. Then χ_S is definable with at most one occurrence of σ.*

Proof. Set $U_S = {\uparrow}S$, $E_S = U_S \setminus S$. Then $S = U_S \setminus {\uparrow}(E_S)$. If $E_S = \varnothing$, then $S = U_S$ is an upset and χ_S is negation-free. Otherwise, $\chi_S = \mathrm{meet}_R(\chi_{U_S}, \sigma(\chi_{\uparrow(E_S)}))$, so one occurrence of σ suffices. $\square$

Lemma 6 (Worst-chain shell identity). *If C is a worst chain for f, then $B_f \cap C$ is exactly the first TRU-block of f on C.*

Proof. Let the first descent on C occur at $c_j \to c_{j+1}$. Then $c_{j+1} \in G_f$, so every point strictly above the first TRU-block lies in ${\uparrow}G_f$ and hence outside B_f. Points below that block are outside A_f, so they are also outside B_f. The first TRU-block itself lies in B_f. $\square$

Lemma 7 (Chain-prefix property). *Let S be an A_f-ideal with $S \subseteq B_f$. Then on every maximal chain C, the set $S \cap C$ is a lower prefix of each TRU-block of f on C.*

Proof. If $x < y$ lie in the same TRU-block on C and $y \in S$, then $y \in B_f$. Lemma 6 implies that x also lies in B_f. Since $x \in A_f$ and S is downward closed inside A_f, it follows that $x \in S$. $\square$

Corollary 3 (No new descents). *Under the hypotheses of Lemma 7, for every maximal chain C,* $\mathrm{desc}(f_S, C) \le \mathrm{desc}(f, C)$.

Proof. Lowering only shortens lower prefixes of existing TRU-blocks, so it cannot create a new $\mathsf{TRU} \to \mathsf{FAL}$ boundary. $\square$

Theorem 7 (Exact one-step decrement for interval shells). *Let S be an A_f-ideal with $W_f \subseteq S \subseteq B_f$. Then* $\mathrm{mcr}(f_S) = \mathrm{mcr}(f) - 1$.

Proof. Fix a worst chain C for f. Since $W_f \subseteq S$, the first TRU-block on C lies in S. Since $S \subseteq B_f$, Lemma 6 forces $S \cap C$ to lie inside that same first block. Hence $S \cap C$ is exactly the first TRU-block, and deleting it lowers the descent count on C by exactly one.

For any other maximal chain D, Corollary 3 gives $\operatorname{desc}(f_S, D) \le \operatorname{desc}(f, D)$. If D is non-worst, then $\operatorname{desc}(f, D) \le \operatorname{mcr}(f) - 1$. If D is worst, the same first-block argument gives $\operatorname{desc}(f_S, D) = \operatorname{mcr}(f) - 1$. Thus all chains have at most $\operatorname{mcr}(f) - 1$ descents after lowering, and at least one worst chain has exactly that many. □

Corollary 4 (Forward interval-shell safety). *Every A_f-ideal S with $W_f \subseteq S \subseteq B_f$ gives a one-step safe shell: it is realized by one occurrence of σ, and lowering along it decreases* mcr *by exactly one.*

Proof. Combine Proposition 11 with Theorem 7. □

Theorem 8 (★ Unconditional containment $W_f \subseteq B_f$). *For every nonconstant, non-monotone $f : R^n \to R$, $W_f \subseteq B_f$.*

Proof. Suppose not: some $w \in W_f$ lies outside B_f. Since $w \in W_f \subseteq A_f$, this means $w \in \uparrow G_f$, so there is a gap point $g \in G_f$ with $g \le w$, and by the definition of G_f a point $a \in A_f$ with $a \le g$, $f(a) = \mathsf{TRU}$, $f(g) = \mathsf{FAL}$. Both containments are strict: $a \ne g$ because $f(a) \ne f(g)$, and $g \ne w$ because $f(g) = \mathsf{FAL}$ while $f(w) = \mathsf{TRU}$. So $a < g < w$.

Since $w \in W_f$, there is a worst chain C for f whose first TRU-block contains w. On C, no descent occurs before the first TRU-block, so all $\operatorname{mcr}(f)$ descents of C occur on the segment of C from w upward.

Build an increasing chain X: ascend to a, then to g, then to w, then follow C upward from w. The segment through $a < g < w$ contributes at least one descent (its f-word passes from TRU at a to FAL at g), and the tail from w contributes $\operatorname{mcr}(f)$ descents. By the refinement lemma (Lemma 10), some maximal chain has at least as many descents as X, so $\operatorname{mcr}(f) \ge \operatorname{desc}(f, X) \ge 1 + \operatorname{mcr}(f)$, a contradiction. □

Corollary 5 (Canonical shell — the interval regime is total). *For every nonconstant, non-monotone f, the A_f-ideal closure of W_f, $S_f = A_f \cap \downarrow W_f$, satisfies $W_f \subseteq S_f \subseteq B_f$. Hence the interval-shell decrement of Corollary 4 applies at every inductive step: every nonconstant non-monotone f admits a one-σ shell whose lowering decreases* mcr *by exactly one.*

Proof. S_f is an A_f-ideal containing W_f by construction. For $S_f \subseteq B_f$, let $x \in S_f$, so $x \in A_f$ and $x \le w$ for some $w \in W_f$. If x were outside B_f, there would be a gap point $g \in G_f$ with $g \le x \le w$, contradicting $w \in B_f$ (Theorem 8). □

D.3 Inductive Completion

Proposition 12 (Monotone base case). *Let $f : R^n \to R$ be nonconstant. If* $\operatorname{mcr}(f) = 0$, *then f is monotone and therefore has a negation-free resolved-face term.*

Proof. If $\operatorname{mcr}(f) = 0$, then no increasing chain contains a descent $\mathsf{TRU} \to \mathsf{FAL}$, so f is monotone. Hence A_f is an upset. Since f is nonconstant, the usual monotone formula $f = \bigvee_{a \in \operatorname{Min}(A_f)} \bigwedge_{i:\, a_i = \mathsf{TRU}} x_i$ computes f using only meet_R and join_R. □

Theorem 9 (Upper bound). *For every nonconstant function $f : R^n \to R$, $\nu(f) \le \operatorname{mcr}(f)$.*

Proof. We induct on $\operatorname{mcr}(f)$. If $\operatorname{mcr}(f) = 0$, Proposition 12 gives $\nu(f) = 0$.

Assume $\operatorname{mcr}(f) > 0$ and the claim already holds for all nonconstant functions of smaller mcr. Since $\operatorname{mcr}(f) > 0$, f is non-monotone, so by Corollary 5 the canonical shell S_f is an A_f-ideal with $W_f \subseteq S_f \subseteq B_f$, and Corollary 4 gives a one-σ lowering step whose residual $f_1 = f_{S_f}$ satisfies $\operatorname{mcr}(f_1) = \operatorname{mcr}(f) - 1$. One occurrence of σ realizes the first step and reconstruction uses only monotone context. Hence $\nu(f) \leq 1 + \nu(f_1)$. By the induction hypothesis, $\nu(f_1) \leq \operatorname{mcr}(f_1) = \operatorname{mcr}(f) - 1$. Therefore $\nu(f) \leq \operatorname{mcr}(f)$. □

Corollary 6 (Exact equality)**.** *For every nonconstant $f : R^n \to R$, $\nu(f) = \operatorname{mcr}(f) = \operatorname{dec}(f)$.*

Proof. The lower bound is Proposition 10 and the upper bound is Theorem 9. Since $\operatorname{mcr}(f) = \operatorname{dec}(f)$ on the two-point chain, all three quantities are equal. □

E Auxiliary Chain-Reversal Tools

This appendix records a small amount of chain notation used as bookkeeping for Appendix D.

E.1 Upward edges and descending edges

Let $X = (x^{(0)} < x^{(1)} < \cdots < x^{(m)})$ be an increasing chain in R^n. Write $\Lambda^{up}(X) = \{(x^{(j)}, x^{(j+1)}) : 0 \leq j < m\}$ for its set of adjacent upward edges. For $f : R^n \to R$, define the descending edge set $\Delta_f(X) = \{(u, v) \in \Lambda^{up}(X) : f(u) > f(v)\}$. Then $\operatorname{mcr}_X(f) := |\Delta_f(X)|$ is the chainwise reversal count, and $\operatorname{mcr}(f) = \max_X \operatorname{mcr}_X(f)$ is the global maximum over increasing chains.

E.2 Ascents and descents in binary chain words

Let $w = (w_0, \ldots, w_m) \in R^{m+1}$ be a binary word. Write $\operatorname{asc}(w)$ for the number of indices j with $w_j = \mathsf{FAL}$ and $w_{j+1} = \mathsf{TRU}$, and $\operatorname{desc}(w)$ for the number of indices j with $w_j = \mathsf{TRU}$ and $w_{j+1} = \mathsf{FAL}$.

Lemma 8. *For every binary word w, $\operatorname{asc}(w) \leq \operatorname{desc}(w) + 1$.*

Proof. Read the word from left to right and decompose it into maximal constant blocks. Every ascent starts a TRU-block, and every descent ends one. At most one TRU-block can fail to be preceded by a descent, namely the initial TRU-block if the word begins at TRU. Hence the number of ascents exceeds the number of descents by at most one. □

Corollary 7. *Let X be an increasing chain and let σ exchange* FAL *and* TRU*. Then $\operatorname{mcr}_X(\sigma \circ f) \leq \operatorname{mcr}_X(f) + 1$.*

Proof. Along the chain X, descents of $\sigma \circ f$ are ascents of the binary word defined by f. Apply Lemma 8. □

E.3 Prefix lowering lemma

Lemma 9. *Fix a binary word along an increasing chain, and lower some* TRU*-entries to* FAL*. If the lowered set is a lower prefix inside each maximal* TRU*-block, then no new descending edge is created.*

Proof. Inside a single maximal TRU-block, lowering a lower prefix changes a block of the form $\mathsf{TRU} \cdots \mathsf{TRU}$ into $\mathsf{FAL} \cdots \mathsf{FAL}\,\mathsf{TRU} \cdots \mathsf{TRU}$. This creates at most one new ascent inside the block and never a new descent. At the block boundaries, the left boundary can only move from $\mathsf{FAL}\,\mathsf{TRU}$ to $\mathsf{FAL}\,\mathsf{FAL}$, and the right boundary can only move from $\mathsf{TRU}\,\mathsf{FAL}$ to either $\mathsf{TRU}\,\mathsf{FAL}$ or $\mathsf{FAL}\,\mathsf{FAL}$. Hence the number of descents does not increase. □

E.4 Subchain refinement lemma

Lemma 10. *Every increasing chain X in R^n extends to a maximal chain C with* $\mathrm{desc}(f, C) \geq \mathrm{mcr}_X(f)$. *Hence the maximum over increasing chains in the definition of* mcr *agrees with the maximum over maximal chains.*

Proof. Refine each upward edge $u < v$ of X into a saturated cover path from u to v and concatenate, extending below the bottom of X and above its top to a maximal chain C. If (u, v) is a descending edge of X, the f-word along the inserted cover path starts at TRU and ends at FAL, so it contains at least one descending cover step. Distinct edges of X refine to disjoint segments of C, so $\mathrm{desc}(f, C) \geq |\Delta_f(X)| = \mathrm{mcr}_X(f)$. □

Remark. Appendix D uses Lemmas 9 and 10 in the interval-shell argument and in Theorem 8. They are recorded separately here only to keep the main exact-cost appendix shorter.

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Authorship & provenance

Won Chul Yang, independent researcher. **This paper is the author's own research**: the mathematical content — the definitions, theorems, and proofs, including the original preprint that this edition revises — is the author's original work. AI assistance (Anthropic Claude family) was used **only for the machine-verification layer**: the Lean 4 mechanization and the exhaustive machine audit/replay of the author's proofs, with the Lean 4 kernel as the acceptance gate for that layer. The interval single-regime simplification (Theorem 8) emerged from that machine audit of the author's Appendix D and was adopted by the author. This division of labor differs from the author's two earlier AI-collaborative releases (`cohomological-price-of-not` and `inversion-wilf-spi`), which were produced in an AI research loop and are labeled accordingly. No claim of academic priority is made beyond full disclosure of how the work was produced. Corrections are invited.